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Example: Lighter molecules like hydrogen and helium have high 'v_{rms}' than heavier molecules such as oxygen and nitrogen at the same temperature.

(ii) Increasing the temperature will increase the r.m.s speed of molecules

We can also write the v_{rms} in terms of gas constant R. Equation (9.18) can be rewritten as follows

$v_{rms} = \sqrt{\frac{3N_A kT}{N_A m}}$ Where N_A is Avogadro number.

Since $N_A k = R$ and $N_A m = M$ (molar mass)

The root mean square speed or r.m.s speed

$$v_{rms} = \sqrt{\frac{3RT}{M}} \qquad (9.19)$$

The equation (9.6) can also be written in terms of rms speed $P = \frac{1}{3} nmv^2_{rms}$

since $v^2_{rms} = \overline{v^2}$

Note Root mean square speed is not the same as average speed. Average speed is 0.92 times of r.m.s speed.

Impact of v_{rms} in nature:

1. **Moon has no atmosphere.**

   The escape speed of gases on the surface of Moon is much less than the root mean square speeds of gases due to low gravity. Due to this all the gases escape from the surface of the Moon.

2. **No hydrogen in Earth's atmosphere**.

   As the root mean square speed of hydrogen is much less than that of nitrogen, it easily escapes from the earth's atmosphere.

   In fact, the presence of nonreactive nitrogen instead of highly combustible hydrogen deters many disastrous consequences.

EXAMPLE 9.2

A room contains oxygen and hydrogen molecules in the ratio 3:1. The temperature of the room is 27°C. The molar mass of 0_2 is 32 g mol^{-1} and for H_2 2 g mol^{-1}. The value of gas constant R is 8.32 J mol^{-1} K^{-1}

Calculate

(a) rms speed of oxygen and hydrogen molecule

(b) Average kinetic energy per oxygen molecule and per hydrogen molecule

(c) Ratio of average kinetic energy of oxygen molecules and hydrogen molecules

Solution

(a) Absolute Temperature
T=27°C =27+273=300 K.

Gas constant R=8.32 J mol^{-1} k^{-1}

For Oxygen molecule: Molar mass M=32 gm=32 x 10^{-3} kg mol^{-1}

rms speed $v_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3\times8.32\times300}{32\times10^{-3}}}$

$= 483.73\ m\ s^{-1} \approx 484\ m\ s^{-1}$

For Hydrogen molecule:

Molar mass M = 2 × 10^{-3} kg mol^{-1}

rms speed $v_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3\times8.32\times300}{2\times10^{-3}}}$

$= 1934\, m\, s^{-1} = 1.93\ k\, m\ \ s^{-1}$

Note that the rms speed is inversely proportional to $\sqrt{M}$ and the molar mass of oxygen is 16 times higher than molar mass of hydrogen. It implies that the rms speed of hydrogen is 4 times greater than rms speed of oxygen at the same temperature. $\frac{1934}{484} \approx 4$.